

AIZA Code-Switching Benchmark Report

Comparing Sahara against two Malagasy-specific fine-tunes on real, self-recorded Malagasy/French code-switched administrative queries (AIZA's actual target domain - see the dataset coverage note below for why this benchmark uses its own audio rather than the official AfriSwitch benchmark).

Models

- **sahara:** `Intron Sahara (voice.intron.io)`
- **whisper-small-malagasy:** `misterkissi/whisper-small-malagasy`
- **whisper-small-waxal-mlg:** `waxal-benchmarking/whisper-small-waxal-mlg`

Summary (lower WER/CER is better)

Model	N	Avg WER	Avg CER	Avg latency (s)
sahara	10	82.6%	45.9%	3.2
whisper-small-malagasy	10	88.2%	48.1%	6.5
whisper-small-waxal-mlg	10	76.4%	46.5%	6.8

Per-utterance results

01 — lost_cin

Reference: Very ny CIN-ko, aiza no manao déclaration de perte?

- **sahara:** 'Ferens Yenne même aux déclarations de perte\n' (WER 80.0%, CER 42.9%)
- **whisper-small-malagasy:** " verin'ny sehy ianinako, aiza ny manoha dia kilarasyon-deo mperita? " (WER 100.0%, CER 53.1%)
- **whisper-small-waxal-mlg:** 'veriny se hianinaako aiza no manao dia kilao rasion deoparita' (WER 80.0%, CER 44.9%)

02 — birth_certificate

Reference: Mila certificat de naissance aho ho an'ny zanako.

- **sahara:** 'La certificat de naissance Wançen à coût\n' (WER 62.5%, CER 37.5%)
- **whisper-small-malagasy:** " na sertifika tena ison-sa ho an'ny zanako. " (WER 62.5%, CER 33.3%)
- **whisper-small-waxal-mlg:** 'lasary tia fikadonisao ho anny zanako' (WER 75.0%, CER 47.9%)

03 — marriage_certificate

Reference: Aiza no maka acte de mariage eto Antananarivo?

- **sahara:** 'Est-ce ma caracte de mariage étant en arrive\n' (WER 87.5%, CER 37.8%)
- **whisper-small-malagasy:** ' haisakaainy makahakita teo maharihazy eto an-tananarivo? ' (WER 100.0%, CER 42.2%)

- **whisper-small-waxal-mlg:** 'aisaka kay no maka haka tato mariazy eto antananarivo' (WER 62.5%, CER 37.8%)

04 — lost_drivers_license

Reference: Very ny permis de conduire-ko, inona no atao?

- **sahara:** 'Permet de conduire Jacques, inamo.\n' (WER 77.8%, CER 53.5%)
- **whisper-small-malagasy:** ' Behery mety ho kanjo irako. Inano atao? ' (WER 88.9%, CER 51.2%)
- **whisper-small-waxal-mlg:** 'tery mety ho gondohirako inona no atao' (WER 66.7%, CER 37.2%)

05 — address_change

Reference: Mila changement d'adresse aho, any amin'ny fokontany ve?

- **sahara:** 'Le changement d'adresse unième fautin fait\n' (WER 87.5%, CER 50.0%)
- **whisper-small-malagasy:** ' loanjy monta adiraisa aho anemy fokitany ve ' (WER 75.0%, CER 50.0%)
- **whisper-small-waxal-mlg:** 'lasandromandady iray sanga any aminny fokotany ve' (WER 75.0%, CER 44.4%)

06 — passport_cost

Reference: Ohatrinona ny frais pour ny passeport vaovao?

- **sahara:** "Au chien n'en ferait pour pas sport fort?\n" (WER 100.0%, CER 50.0%)
- **whisper-small-malagasy:** ' ho tsy ny fray pora-pasiparva vao? ' (WER 100.0%, CER 47.7%)
- **whisper-small-waxal-mlg:** 'ohatrany ny freimpasimbaovao vaovao' (WER 71.4%, CER 47.7%)

07 — expired_cin

Reference: Tsy mety ny carte d'identité-ko, tapitra ny validité.

- **sahara:** 'Si méthan cachent d'identité accout obchant validité.\n' (WER 88.9%, CER 41.2%)
- **whisper-small-malagasy:** ' tsy mety ny karita didon-titehako tapitra ny validy ite. ' (WER 55.6%, CER 19.6%)
- **whisper-small-waxal-mlg:** 'tsy mety ny karita dido antitehako tapitra ny validite' (WER 33.3%, CER 17.6%)

08 — land_title

Reference: Manao ahoana ny procédure ho an'ny titre foncier?

- **sahara:** 'La procédure un titre français?\n' (WER 75.0%, CER 54.2%)
- **whisper-small-malagasy:** ' mprosesy hirao anehitra firan-fonsiegle, ' (WER 100.0%, CER 75.0%)
- **whisper-small-waxal-mlg:** 'posedure ho anny hita frantsay fotsy iaka e' (WER 100.0%, CER 77.1%)

09 — criminal_record

Reference: Mila extrait de casier judiciaire aho, aiza no mankany?

- **sahara:** 'Mais il a extrait de gaz judiciaires Wacen Mcane?\n' (WER 100.0%, CER 45.3%)
- **whisper-small-malagasy:** " milaheksitireto eo gazehajodisy herao aizakaian'ny makane? " (WER 100.0%, CER 60.4%)

- **whisper-small-waxal-mlg:** 'milaiky sy reto ka azy avy ity sery aho aiza kay no mankany' (WER 100.0%, CER 56.6%)

10 — processing_time

Reference: Firy andro no ilaina hahazoana ny déclaration de perte?

- **sahara:** 'C'est rianoz à une déclaration de perte.\n' (WER 66.7%, CER 46.3%)
- **whisper-small-malagasy:** ' firiandry noho aza hoana diaekilarasy ao teo perta? ' (WER 100.0%, CER 48.1%)
- **whisper-small-waxal-mlg:** 'fy rihandro no azo ahoana dia kilao soa teo mpiata' (WER 100.0%, CER 53.7%)

Dataset coverage note

Sahara's published language list (confirmed via docs.voice.intron.io and the challenge's own WhatsApp announcement) does not include Malagasy. The official `intronhealth/AfriSwitch` benchmark - 14 African languages code-switched with English - also does not include Malagasy, and the dataset itself is manually gated with no public preview, so we could not rely on approval landing before the deadline. No public code-switched Malagasy/French benchmark dataset appears to exist yet. That gap is itself a finding, not just a limitation: it's why AIZA always shows the raw transcript rather than silently 'correcting' it, and always offers a typed fallback instead of guessing when confidence is low.